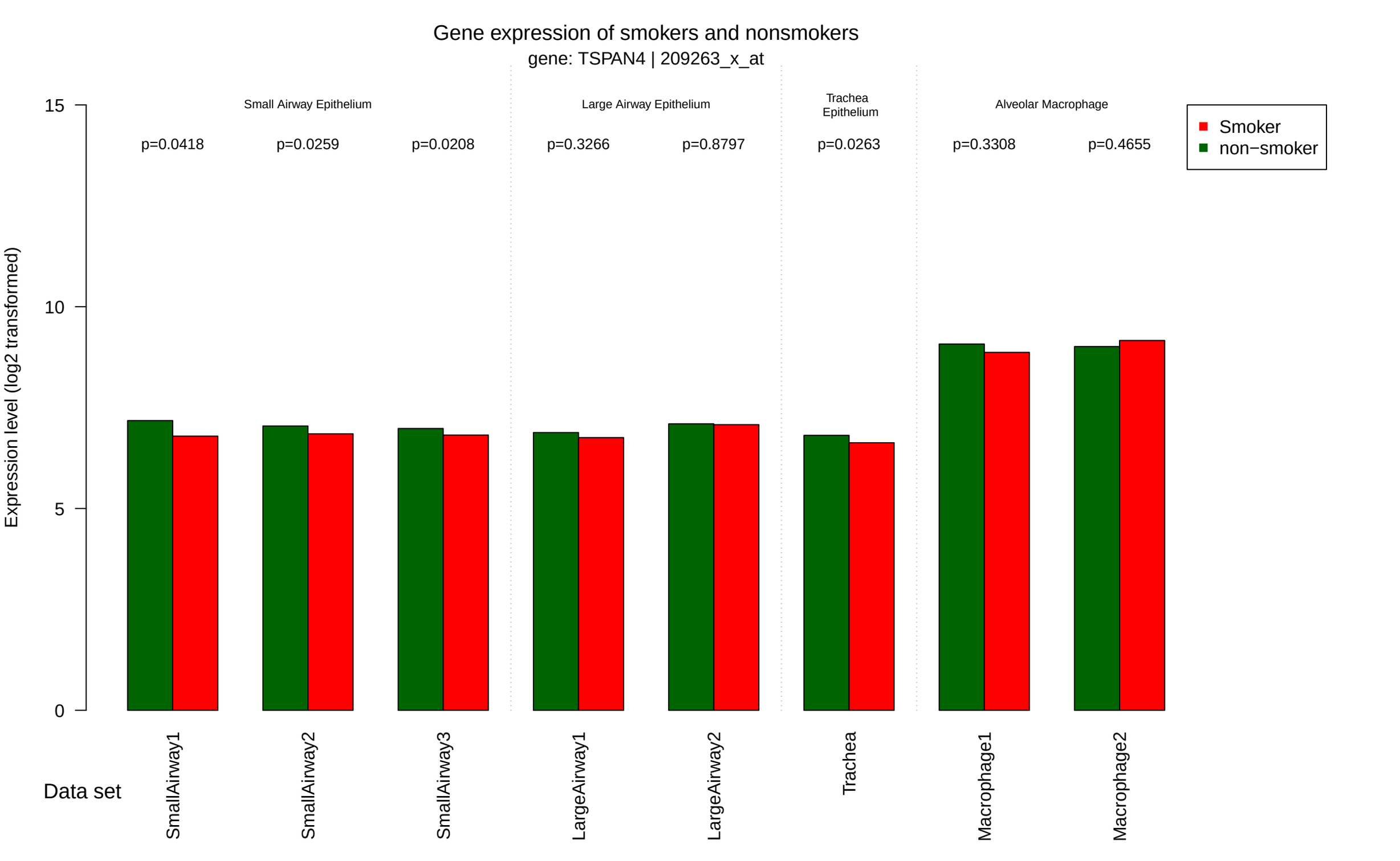
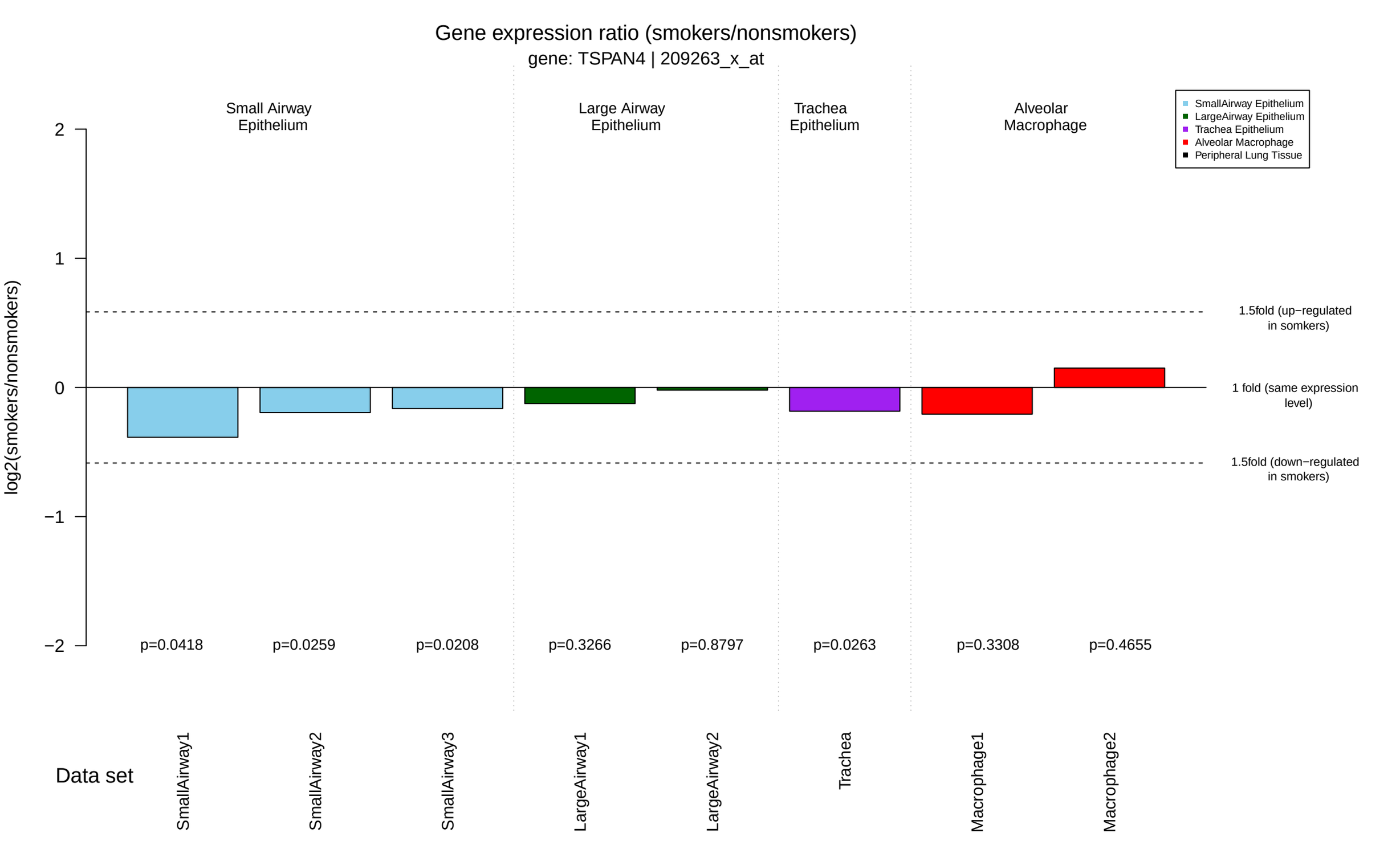
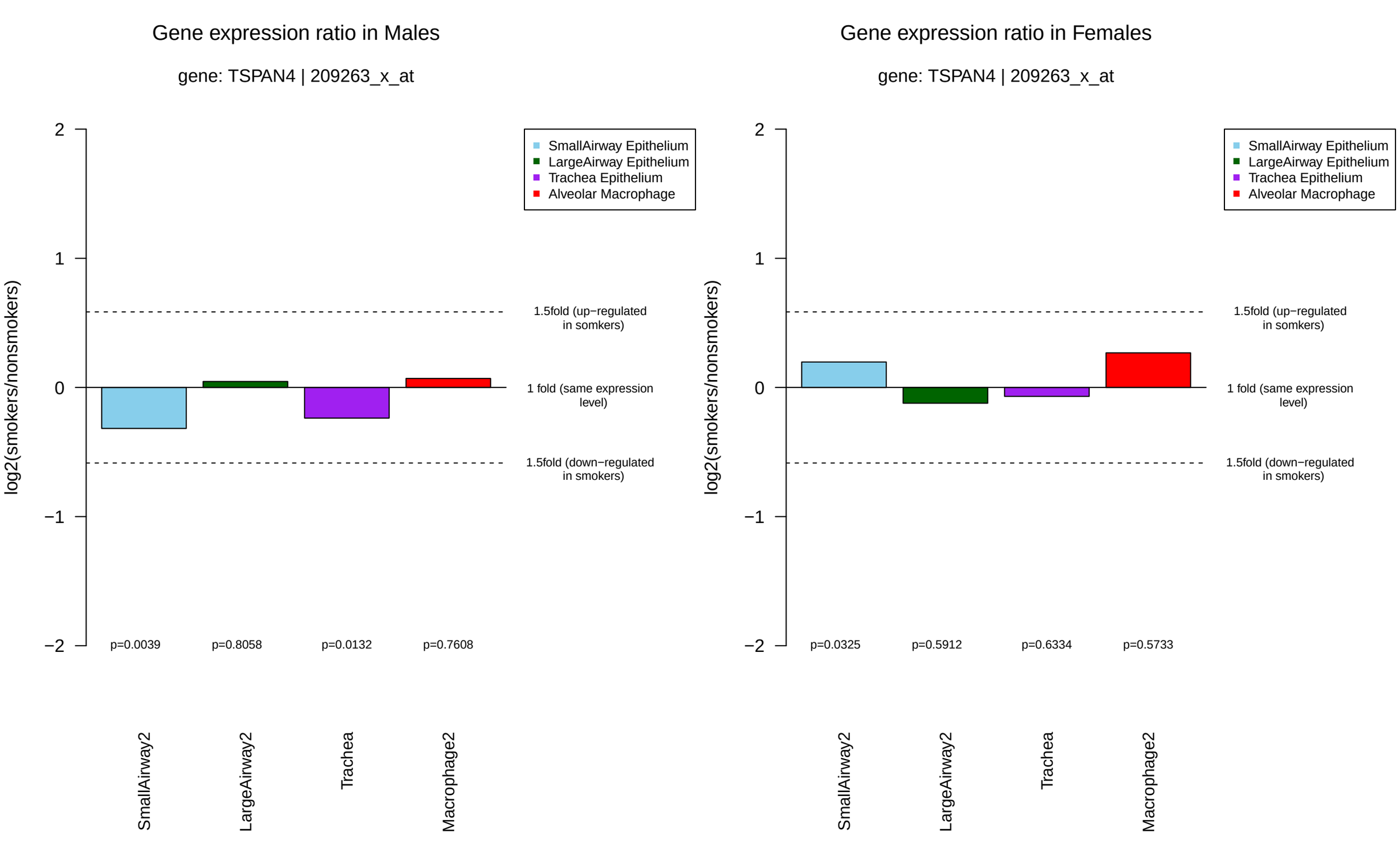
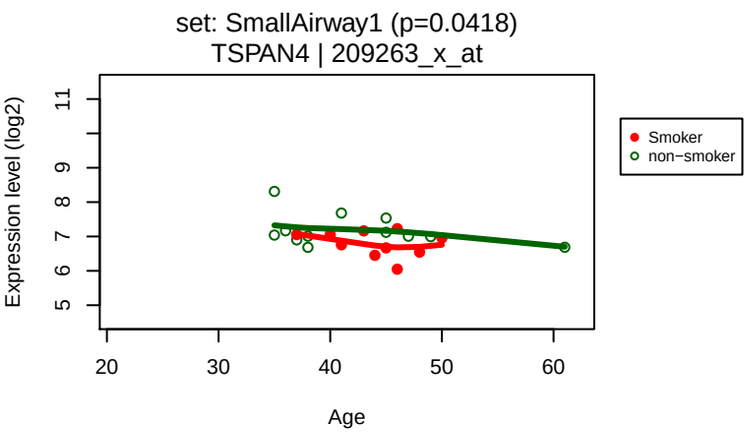
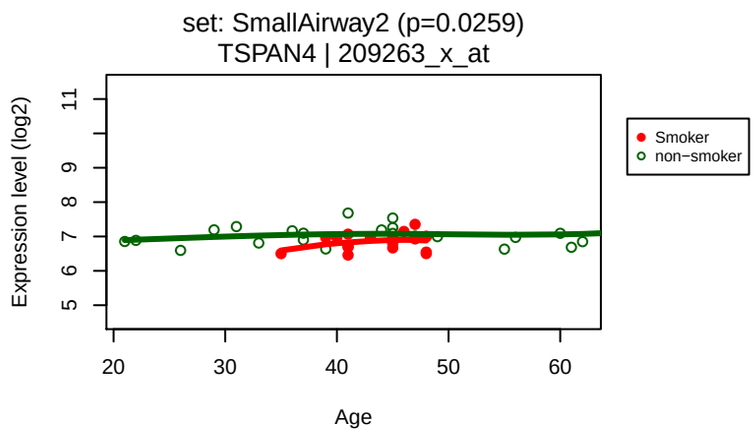
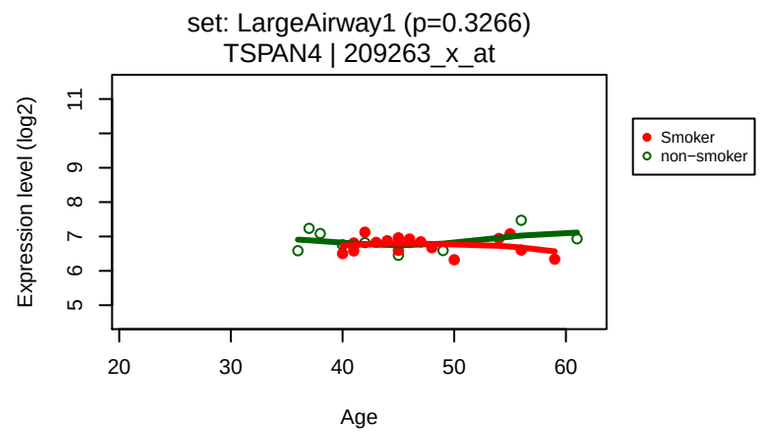

Gene expression of smokers and nonsmokers
gene: TSPAN4 | 209263_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0418
p=0.0259
p=0.0208
p=0.3266
p=0.8797
p=0.0263
p=0.3308
p=0.4655
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

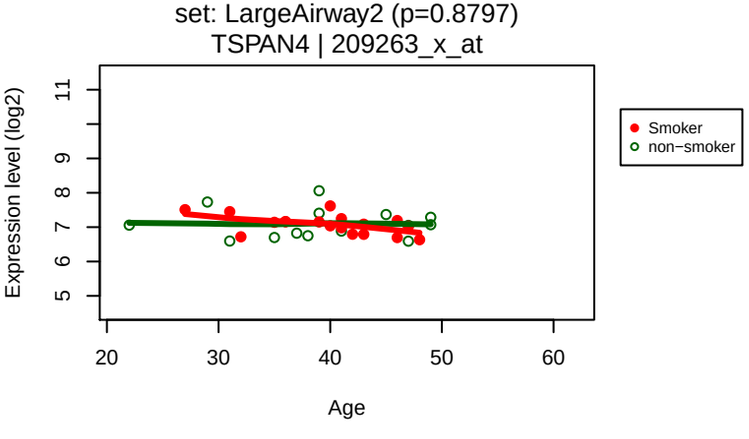

Gene expression ratio (smokers/nonsmokers)
gene: TSPAN4 | 209263_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0418
p=0.0259
p=0.0208
p=0.3266
p=0.8797
p=0.0263
p=0.3308
p=0.4655
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

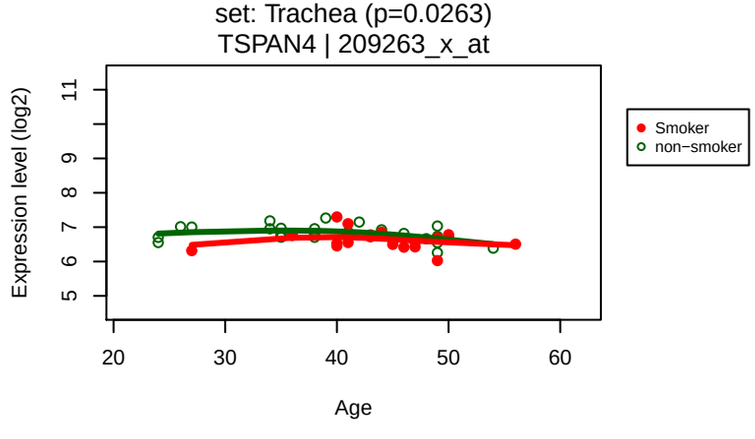

Gene expression ratio in Males
gene: TSPAN4 | 209263_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0039
p=0.8058
p=0.0132
p=0.7608
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TSPAN4 | 209263_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0325
p=0.5912
p=0.6334
p=0.5733
SmallAirway2
LargeAirway2
Trachea
Macrophage2

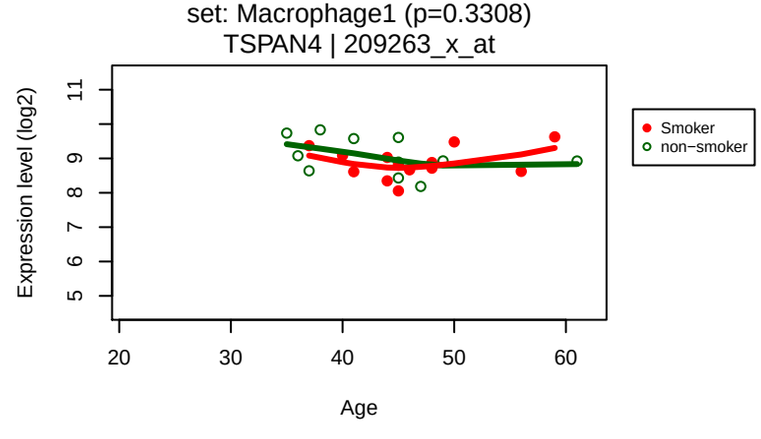

set: SmallAirway1 (p=0.0418)
TSPAN4 | 209263_x_at
Expression level (log2)
Age
Smoker
non−smoker

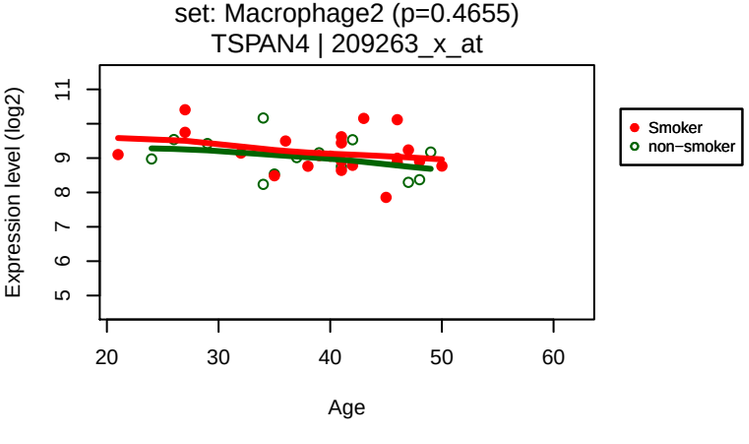

set: SmallAirway2 (p=0.0259)
TSPAN4 | 209263_x_at
Expression level (log2)
Age
Smoker
non−smoker

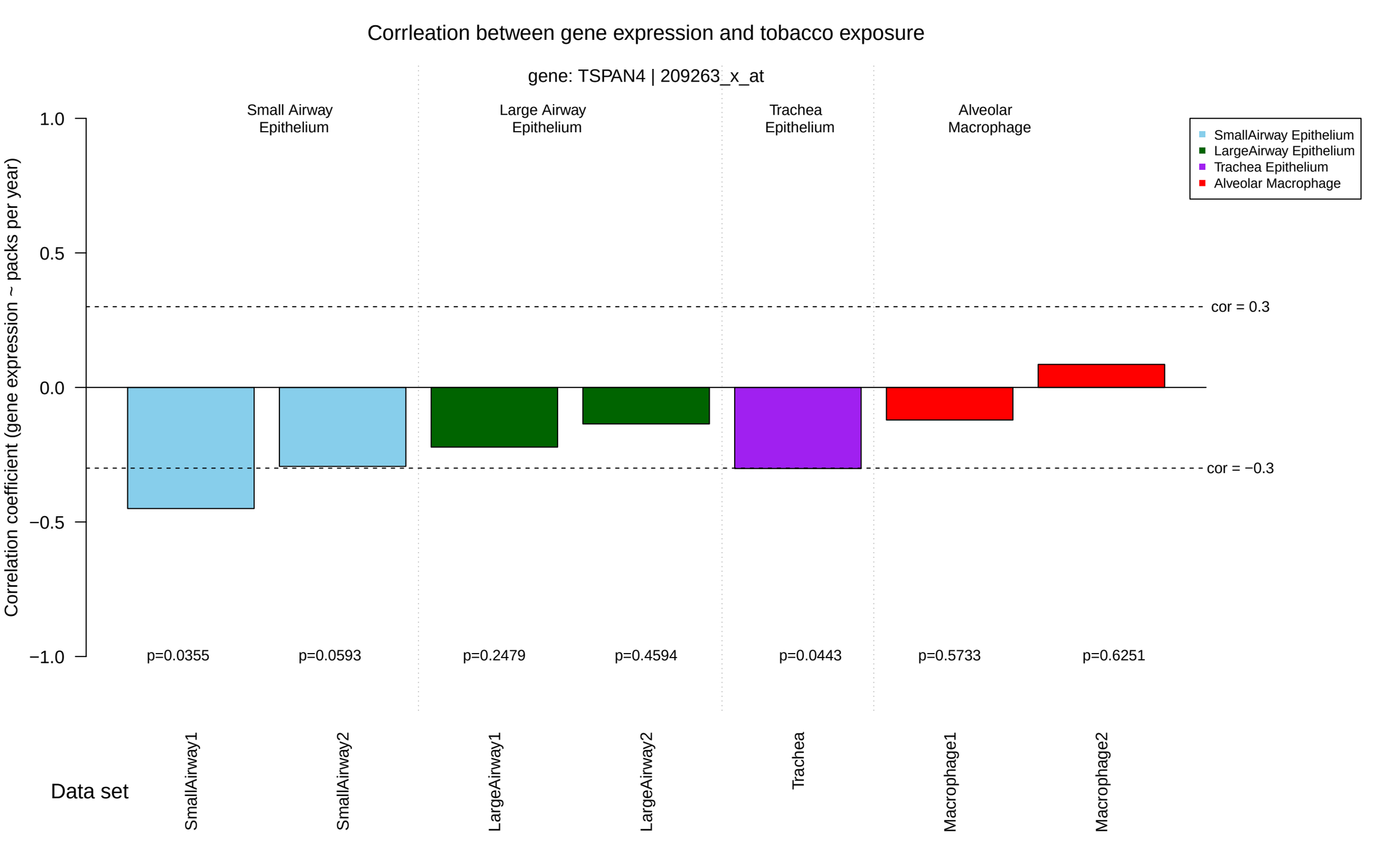

set: LargeAirway1 (p=0.3266)
TSPAN4 | 209263_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8797)
TSPAN4 | 209263_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0263)
TSPAN4 | 209263_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3308)
TSPAN4 | 209263_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4655)
TSPAN4 | 209263_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TSPAN4 | 209263_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0355
p=0.0593
p=0.2479
p=0.4594
p=0.0443
p=0.5733
p=0.6251
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2